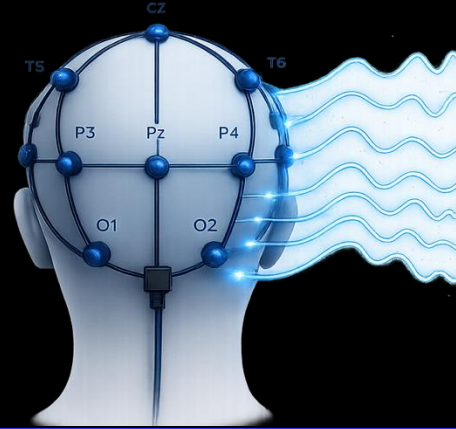


UC San Diego: Cogs 138 Neural Data Science
Final Project Presentation



From Personal SSVEP EEG to Public Benchmark Decoding

[Project Report](#) · [Project Repo](#)



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Decoding Visual Attention from SSVEP EEG

Question:

Can frequency-locked EEG decode the visual target a person attends?

Pipeline:

measurement >> preprocessing >> representation >> modeling >> inference >> action

Privacy note: my personal OpenBCI recordings were used as a pilot measurement system, but raw personal EEG data are excluded from the public repo.

Key EDA Figure

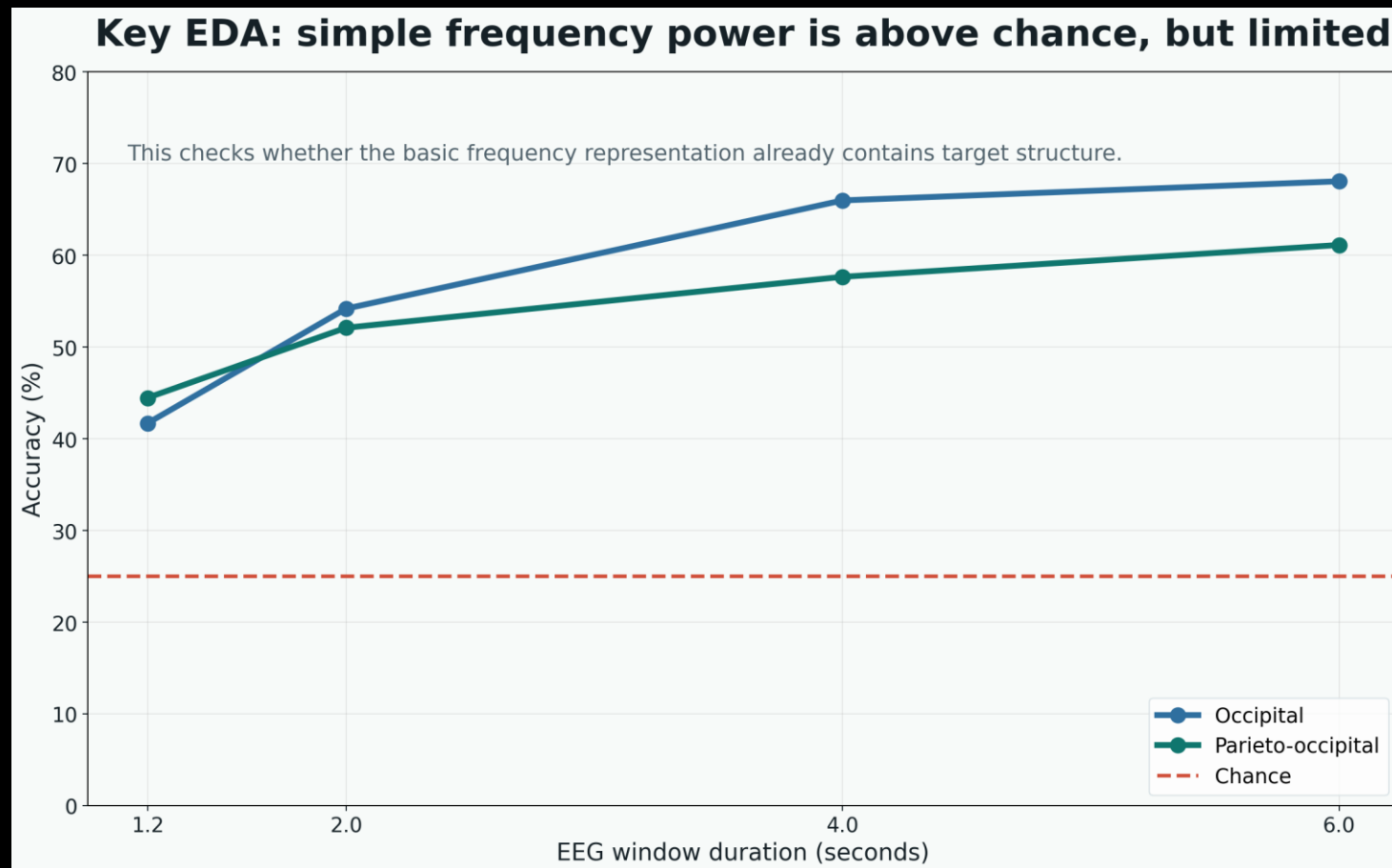
Simple target-frequency power is above chance, but limited

Public Wang2016 validation

Subjects 1-6

Four matched targets: 8, 10, 12, 15 Hz

Chance = 25%



Key Result

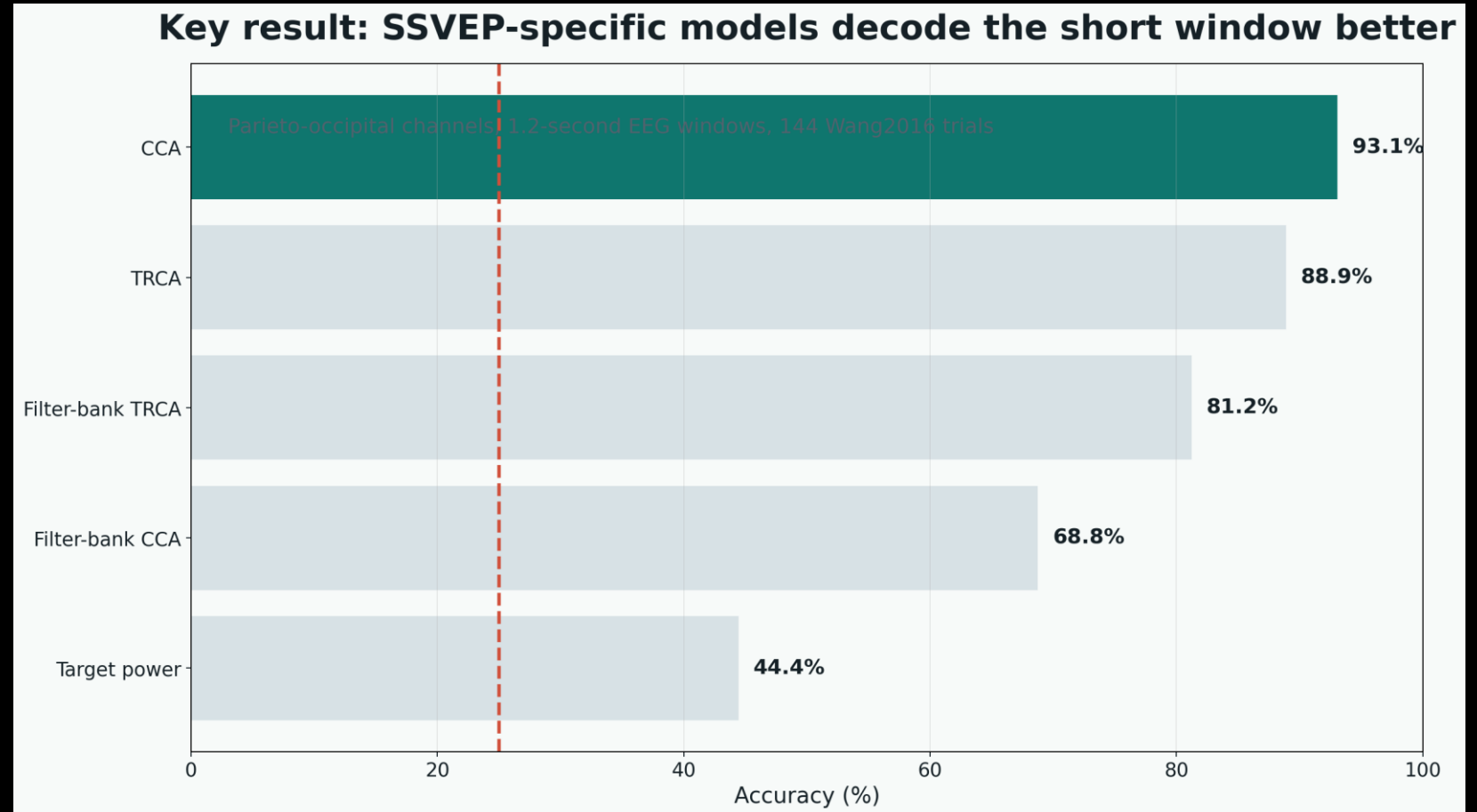
Model choice matters at the
1.2-second window

CCA, parieto-occipital channels

134 / 144 correct

93.1% accuracy

Permutation $p = 0.0002$



Takeaway

- Target-frequency power detects some structure
- CCA and TRCA work better because they match SSVEP physiology
- The useful neural signal is frequency-locked visual cortical activity
- Representation and model choice determine whether EEG becomes actionable